
XLM-RoBERTa

Unsupervised Cross-Lingual Representation Learning at Scale

A Comprehensive Technical and Academic Report

*Architecture, Pretraining, Evaluation, and Applications of a
State-of-the-Art Multilingual Language Model*

Model Family: XLM-R (Base / Large)

Developed by: Meta AI (Facebook AI Research, FAIR)

Original Paper: *Unsupervised Cross-lingual Representation
Learning at Scale* (Conneau et al., 2019/2020)

Languages Covered: 100

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Abstract

This report presents a comprehensive academic overview of **XLM-RoBERTa (XLM-R)**, a multilingual masked-language model introduced by researchers at Meta AI (Facebook AI Research) in the 2019 paper “*Unsupervised Cross-lingual Representation Learning at Scale*.” The report traces the historical motivation behind cross-lingual representation learning, details the Transformer encoder architecture on which XLM-R is built, and explains its pretraining procedure using the Masked Language Modeling (MLM) objective over the CC-100 multilingual corpus. It further describes the SentencePiece tokenization scheme, contrasts the Base and Large model configurations, and evaluates the model’s advantages and limitations relative to BERT, RoBERTa, multilingual BERT (mBERT), and the original XLM model. A dedicated chapter summarizes the original research paper in full academic form – motivation, methodology, experimental design, results, and contributions – with key figures and tables redrawn for clarity. Practical applications are discussed, including cross-lingual natural language inference, question answering, named-entity recognition, and low-resource text-classification tasks such as **Bangla fake-news detection**. The report closes with IEEE-style references and clickable hyperlinks to the original paper, model repositories, and documentation, making it suitable as a self-contained reference for students and researchers studying multilingual natural language processing.

Keywords – XLM-RoBERTa, Transformer, Masked Language Modeling, Cross-Lingual Transfer, CC-100, SentencePiece, mBERT, XNLI, MLQA, Named Entity Recognition, Bangla NLP.

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Chapter 1 Introduction to XLM-RoBERTa

1.1 What is XLM-RoBERTa?

XLM-RoBERTa (commonly abbreviated **XLM-R**) is a large-scale, multilingual, Transformer-based language model designed to produce high-quality contextual representations for text written in **100 different languages** using a single shared set of parameters. It belongs to the family of *masked language models* (MLMs), meaning that during pretraining it learns to predict deliberately hidden (“masked”) tokens within a sentence using only the surrounding context. Because it is trained purely on unlabeled monolingual text drawn from the web, XLM-R requires no parallel or translated corpora, which allows it to scale to languages for which parallel data is scarce or unavailable.

Architecturally, XLM-R is an encoder-only Transformer that closely follows the design of RoBERTa [4], which itself is a robustly optimized variant of BERT [3]. The key innovation of XLM-R is not a new neural architecture, but rather the demonstration that a single multilingual model, trained at sufficient scale on sufficiently large and clean multilingual data, can match or exceed the performance of dedicated monolingual models while simultaneously supporting cross-lingual transfer – the ability to fine-tune a model on labeled data in one language (typically English) and apply it directly to unseen languages with little or no additional supervision.

1.2 Why This Report Was Written

This report is intended as a self-contained academic reference document. It consolidates the theoretical foundations, training methodology, empirical results, and practical applications of XLM-R into a single structured resource, suitable for coursework, literature review sections of a thesis, or general study of multilingual Natural Language Processing (NLP). Every major design decision behind XLM-R – from its choice of tokenizer to its training corpus – is explained together with the reasoning that motivated it.

1.3 Document Roadmap

Chapter 2 discusses the historical context and motivation for building XLM-R. Chapter 3 identifies the research organization responsible for the model. Chapters 4–7 describe the underlying Transformer architecture, the masked-language-modeling pretraining objective, the CC-100 dataset, and SentencePiece tokenization. Chapter 8 compares the Base and Large model configurations. Chapter 9 evaluates advantages and limitations, while Chapter 10 benchmarks XLM-R against BERT, RoBERTa, mBERT, and XLM. Chapter 11 surveys real-world applications, including Bangla fake-news detection. Chapter 12 reports empirical performance on XNLI, MLQA, and NER. Chapter 13 provides a full summary of the original research paper, and Chapter 14 concludes the report, followed by IEEE-style references.

Chapter 2 History and Motivation

2.1 The Road to Cross-Lingual Language Models

The development of XLM-R sits at the end of a rapid, multi-year evolution in NLP representation learning:

- **2017 – The Transformer.** Vaswani et al. introduced the Transformer architecture in “*Attention Is All You Need*” [6], replacing recurrence with self-attention and enabling far more parallelizable and scalable training.
- **2018 – BERT.** Devlin et al. proposed BERT [3], which used a bidirectional Transformer encoder pretrained with Masked Language Modeling and Next Sentence Prediction, achieving state-of-the-art results on numerous English NLU benchmarks.
- **2019 – Multilingual BERT (mBERT).** A multilingual variant of BERT was released, trained jointly on Wikipedia text from over 100 languages using a shared vocabulary, offering the first widely used cross-lingual Transformer encoder – though trained on comparatively small, imbalanced Wikipedia data.
- **2019 – XLM.** Lample and Conneau [5] proposed cross-lingual pretraining objectives, including Translation Language Modeling (TLM), which uses parallel sentence pairs to explicitly encourage cross-lingual alignment.
- **2019 – RoBERTa.** Liu et al. [4] showed that BERT was significantly undertrained, and that removing the Next Sentence Prediction objective, training with larger batches, more data, and longer schedules substantially improved performance.
- **2019 – XLM-RoBERTa.** Conneau et al. [1] combined the RoBERTa training recipe with a massively multilingual, web-scale corpus (CC-100) covering 100 languages, showing that scaling data and model capacity – rather than adding cross-lingual supervision – was the key to strong cross-lingual transfer.

2.2 Why XLM-R Was Developed

Prior multilingual models, especially mBERT and the original XLM, were limited by the relatively small and imbalanced amount of pretraining data available for most of the world’s languages – primarily Wikipedia, which is small or nearly absent for many low-resource languages. The authors of XLM-R identified several concrete motivations for a new model:

1. **Data scarcity for low-resource languages.** Wikipedia-based corpora used by mBERT and XLM-100 are extremely small for the majority of languages, undermining representation quality.

2. **Undertraining of prior multilingual models.** Following the RoBERTa finding that BERT was undertrained, the authors hypothesized that multilingual models could similarly benefit from more data, larger models, and longer training.
3. **The “curse of multilinguality.”** Adding more languages to a fixed-capacity model dilutes per-language capacity; the authors wanted to systematically study this trade-off and show how to mitigate it through increased model capacity.
4. **Removing the need for parallel data.** Unlike XLM’s Translation Language Modeling objective, which requires parallel corpora that do not exist at scale for most language pairs, XLM-R relies solely on the MLM objective over monolingual data, making it far easier to scale to 100 languages.
5. **Bridging the monolingual/multilingual performance gap.** Earlier multilingual models sacrificed per-language performance compared to dedicated monolingual models. The authors aimed to show, for the first time, that a single multilingual model could remain competitive with strong monolingual baselines such as RoBERTa on English benchmarks (e.g., GLUE) while excelling at cross-lingual tasks.

Core Idea: XLM-R was developed to test – and ultimately confirm – the hypothesis that *scale* (more data, more languages, more parameters, longer training) is sufficient to produce strong, general-purpose cross-lingual representations, without requiring cross-lingual supervision signals such as parallel corpora or dictionaries.

Chapter 3 Who Created XLM-RoBERTa

3.1 Meta AI / Facebook AI Research (FAIR)

XLM-RoBERTa was developed by researchers at **Facebook AI Research (FAIR)**, the artificial intelligence research division of Facebook, Inc. – the organization now operating under the corporate name **Meta Platforms, Inc.** (“Meta AI”). The original paper, *Unsupervised Cross-lingual Representation Learning at Scale*, was authored by Alexis Conneau, Kartikay Khandelwal, Naman Goyal, Vishrav Chaudhary, Guillaume Wenzek, Francisco Guzmán, Edouard Grave, Myle Ott, Luke Zettlemoyer, and Veselin Stoyanov [1].

3.2 Publication Timeline

The paper was first submitted to arXiv on **5 November 2019** and later revised (v2) on **8 April 2020**. It was subsequently published at the **58th Annual Meeting of the Association for Computational Linguistics (ACL 2020)** [2]. FAIR released the pretrained model weights, tokenizer, and training code publicly, and the model was rapidly integrated into the Hugging Face **transformers** library, which greatly accelerated its adoption across the NLP community.

3.3 Relationship to Other Meta AI Research

XLM-R builds directly on two other Meta AI / FAIR research lines:

- **RoBERTa** [4], also developed at FAIR, which established the optimized training recipe (dynamic masking, larger batches, no Next Sentence Prediction, longer training) that XLM-R adopts.
- **XLM** [5], an earlier FAIR project on cross-lingual language modeling that established the use of Masked Language Modeling and Translation Language Modeling for cross-lingual transfer.

FAIR later extended this line of work with **XLM-R XL and XXL** [8], which scale the model to 3.5 billion and 10.7 billion parameters respectively, further improving cross-lingual benchmark performance.

Chapter 4 The Transformer Architecture

4.1 Overview

XLM-R uses the **encoder** half of the original Transformer architecture proposed by Vaswani et al. [6]. Unlike recurrent neural networks, the Transformer processes an entire sequence in parallel using a mechanism called *self-attention*, which allows every token to directly attend to every other token in the sequence regardless of distance. This removes the sequential bottleneck of RNNs/LSTMs and enables far more efficient training on modern hardware.

4.2 Core Components

Each Transformer encoder layer used in XLM-R consists of two principal sub-layers, each wrapped with a residual connection and layer normalization:

1. **Multi-Head Self-Attention (MHSA).** For each token, the model computes Query (Q), Key (K), and Value (V) vectors. Attention scores are computed as

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V,$$

where d_k is the dimensionality of the key vectors. Multiple attention “heads” run in parallel, allowing the model to jointly attend to information from different representation subspaces at different positions.

2. **Position-wise Feed-Forward Network (FFN).** A two-layer fully connected network with a non-linear activation (GELU) is applied identically and independently to each token position:

$$\text{FFN}(x) = \max(0, xW_1 + b_1)W_2 + b_2.$$

Because self-attention has no inherent notion of token order, **positional embeddings** are added to the input token embeddings so the model can make use of sequence order.

4.3 Encoder Stack Diagram

Figure 4.1 illustrates one Transformer encoder block as used in XLM-R, along with the overall stacked-encoder design (redrawn based on the standard Transformer/BERT-style encoder description; not copied from any source figure).

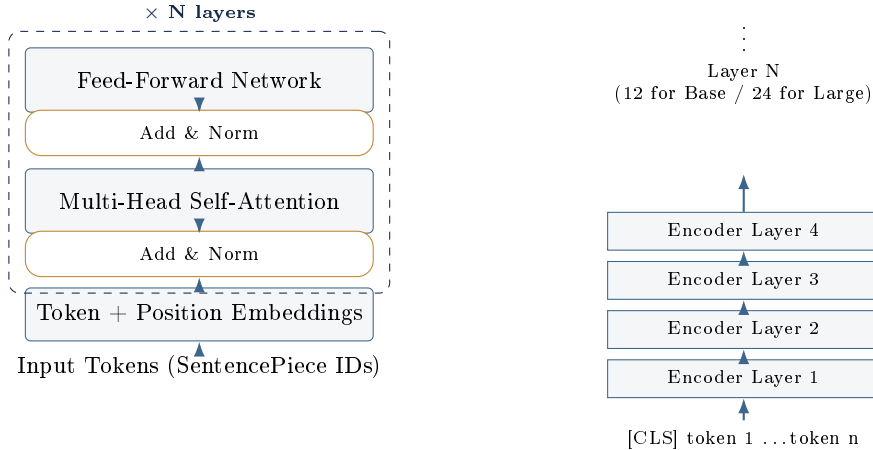


Figure 4.1: Left: a single Transformer encoder block (self-attention + feed-forward, each with residual Add & Norm), as used in XLM-R. Right: the full encoder stack of N such layers, with contextual output representations for each token. Diagram redrawn for this report; not reproduced from the original publication.

4.4 Why an Encoder-Only Design?

XLM-R (like BERT and RoBERTa) uses only the Transformer *encoder*, not the decoder. This is well suited to *understanding* tasks – classification, sequence labeling, extractive question answering – where the goal is to produce a rich contextual representation of an input sequence rather than to autoregressively generate new text. Because self-attention in the encoder is *bidirectional* (every token can see every other

token, including those to its right), the model builds deeply contextualized representations that leverage the full sentence context in both directions.

Chapter 5 Pretraining Process: Masked Language Modeling

5.1 The Masked Language Modeling Objective

XLM-R is pretrained using a single, purely self-supervised objective: **Masked Language Modeling (MLM)**, following the RoBERTa recipe rather than the original BERT recipe (which also used Next Sentence Prediction, later shown to be unnecessary). During pretraining:

1. A stream of text is sampled from the multilingual corpus (CC-100), with languages sampled according to a smoothed distribution (described in Chapter 6) rather than in proportion to raw corpus size.
2. Approximately **15% of SentencePiece subword tokens** in each sequence are selected for masking.
3. Of the selected tokens: 80% are replaced with a special [MASK] token, 10% are replaced with a random token from the vocabulary, and 10% are left unchanged – the same masking recipe introduced by BERT.
4. The model must predict the original identity of each masked token using only the surrounding bidirectional context, minimizing a cross-entropy loss over the vocabulary.

5.2 Dynamic Masking

Following RoBERTa, XLM-R uses **dynamic masking**: the masking pattern is generated freshly each time a sequence is fed to the model during training, rather than being fixed once during data preprocessing (as in the original BERT). This exposes the model to many different masking patterns of the same underlying text over the course of training, improving robustness.

5.3 Multilingual Sampling

Because the amount of text differs by several orders of magnitude across languages (e.g., English versus low-resource languages), sampling text purely in proportion to corpus size would starve low-resource languages of training exposure and cause the model to overfit to high-resource languages. XLM-R instead samples sequences with a probability

$$p_l \propto \left(\frac{n_l}{\sum_i n_i} \right)^\alpha,$$

where n_l is the amount of data for language l and $\alpha < 1$ (specifically $\alpha = 0.3$ in the original paper) is a smoothing exponent that up-samples low-resource languages and down-samples high-resource languages, balancing the trade-off between per-language data volume and representation quality across all 100 languages.

5.4 No Cross-Lingual Supervision Required

A key distinction from the original XLM model is that XLM-R **does not** use the Translation Language Modeling (TLM) objective, which required parallel (translated) sentence pairs. XLM-R relies exclusively on monolingual MLM, demonstrating that explicit cross-lingual supervision is not required to obtain strong cross-lingual transfer – the shared subword vocabulary and joint multilingual training are sufficient to induce a shared cross-lingual representation space.

Chapter 6 The CC-100 Dataset

6.1 Motivation for a New Corpus

Prior multilingual models such as mBERT and XLM-100 were trained on Wikipedia dumps, which are small and highly imbalanced across languages – abundant for English, French, or German, but extremely sparse

for many low-resource languages. To overcome this bottleneck, the XLM-R authors constructed a new corpus, **CC-100**, built by extracting and filtering monolingual text for 100 languages from the **Common Crawl** web archive, using the CCNet pipeline for large-scale data collection and quality filtering.

6.2 Construction Pipeline

The CC-100 corpus was built through the following stages:

1. **Web crawling.** Raw web pages are sourced from Common Crawl snapshots.
2. **Language identification.** A fastText-based language classifier assigns each document to one of 100 target languages.
3. **Deduplication.** Near-duplicate paragraphs are removed to reduce redundancy and memorization.
4. **Quality filtering.** A language-model-based filter (trained on Wikipedia) scores documents and discards low-quality or non-natural-language text (e.g., boilerplate, spam), keeping only text whose perplexity resembles that of clean encyclopedic text.

6.3 Scale of the Corpus

The resulting corpus totals approximately **2.5 terabytes** of filtered text spanning 100 languages – several orders of magnitude larger than the Wikipedia-based corpora used by mBERT and XLM-100, especially for lower-resource languages. Figure 6.1 redraws, at a conceptual level, the relative scale gain reported in the original paper for a sample of languages spanning high- to low-resource categories.

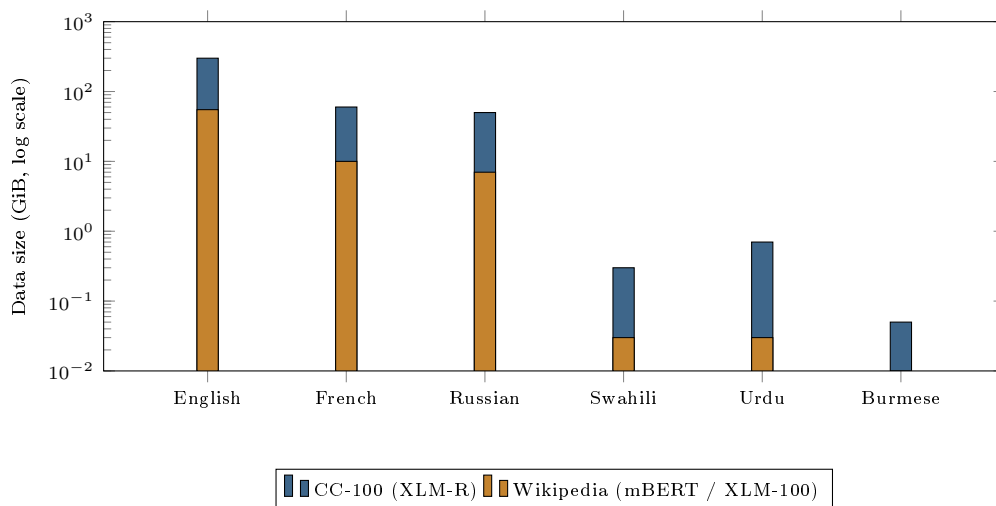


Figure 6.1: Conceptual illustration (redrawn, illustrative values) of the data-scale advantage of CC-100 over Wikipedia-based corpora, especially for low-resource languages such as Swahili and Burmese. The original paper reports this effect for 88 shared languages on a log scale; exact per-language values are approximate here for illustration.

6.4 Impact on Low-Resource Languages

Because CC-100 dramatically increases available monolingual data for low-resource languages relative to Wikipedia, XLM-R shows especially large gains on those languages – for example, the original paper reports XNLI accuracy improvements of **+15.7 points for Swahili** and **+11.4 points for Urdu** over the earlier XLM model, a direct consequence of the improved corpus.

Chapter 7 Tokenization: SentencePiece

7.1 Why a Shared Subword Tokenizer?

Supporting 100 languages – many with different scripts, morphological structures, and no natural word boundaries (e.g., Chinese, Japanese, Thai) – rules out simple whitespace or word-level tokenization. XLM-R therefore uses **SentencePiece** [7], a language-agnostic subword tokenizer that operates directly on raw Unicode text without requiring pre-tokenization or language-specific rules.

7.2 How SentencePiece Works

- SentencePiece treats the input as a raw stream of Unicode characters (including whitespace, which is encoded as a special meta-symbol, typically “_”), avoiding any dependency on language-specific word-segmentation tools.
- It learns a fixed-size subword vocabulary using a **unigram language model** algorithm, selecting the subword units that best compress the training corpus under a probabilistic unigram model.
- Because tokenization and detokenization are fully reversible, the same tokenizer can process any of the 100 target languages consistently, including languages without spaces between words.

7.3 XLM-R’s Vocabulary

XLM-R uses a shared SentencePiece vocabulary of **250,000 tokens** trained on the multilingual CC-100 corpus – substantially larger than mBERT’s WordPiece vocabulary of about 110,000 tokens – allowing better lexical coverage across many scripts while limiting excessive fragmentation of rare words. XLM-R uses a *full softmax* over this large vocabulary during MLM pretraining, in contrast to some other multilingual models that use sampled or adaptive softmax approximations to reduce computation.

Key takeaway: SentencePiece’s language-agnostic, character-level approach is what allows a single XLM-R tokenizer – and therefore a single shared vocabulary and embedding matrix – to serve 100 typologically diverse languages without any per-language tokenization logic.

Chapter 8 Model Variants: Base vs. Large

8.1 Configuration Comparison

The original paper releases two primary model sizes, summarized in Table 8.1. Later work by the same research group [8] extended the family with XL (3.5B parameters) and XXL (10.7B parameters) variants, though Base and Large remain the most widely used in practice.

Table 8.1: XLM-R Base vs. Large architecture configurations.

Configuration	XLM-R Base	XLM-R Large
Transformer layers (L)	12	24
Hidden size (H)	768	1024
Attention heads (A)	12	16
Feed-forward dimension	3072	4096
Vocabulary size	250,000	250,000
Total parameters	≈270 million	≈550 million
Pretraining data	2.5 TB filtered CC-100 (100 languages)	
Pretraining objective	Masked Language Modeling (MLM)	

8.2 Practical Trade-offs

- **XLM-R Base** offers a favorable balance of accuracy and computational cost, making it suitable for fine-tuning with limited GPU memory and for latency-sensitive applications.
- **XLM-R Large** achieves consistently higher accuracy on cross-lingual benchmarks (see Chapter 12) at roughly double the parameter count and proportionally higher training/inference cost.

- Both variants share the same 250K-token vocabulary and tokenizer, so downstream code can switch between them with minimal modification.
- The later **XL/XXL** variants further increase capacity to mitigate the “curse of multilinguality,” improving low-resource-language performance at substantially higher computational cost.

Chapter 9 Advantages and Limitations

9.1 Advantages

- **Broad language coverage.** A single model supports 100 languages with one shared architecture and vocabulary, removing the need to train and maintain separate monolingual models.
- **Strong zero-shot cross-lingual transfer.** A classifier fine-tuned only on English labeled data can be applied directly to other languages with substantially better accuracy than earlier multilingual models.
- **No parallel data required.** Because pretraining uses only monolingual MLM, XLM-R can be trained (and extended) using easily obtainable web text rather than scarce parallel corpora.
- **Competitive with monolingual models.** XLM-R Large is shown to be competitive with monolingual RoBERTa on English GLUE tasks, despite being trained on 100 languages simultaneously – a result the original authors highlight as demonstrating that multilingual training need not sacrifice per-language quality at sufficient scale.
- **Large, high-quality corpus.** The CC-100 dataset provides orders of magnitude more training data than Wikipedia-based alternatives, especially benefiting low-resource languages.
- **Wide ecosystem support.** Public release of weights and integration into the Hugging Face `transformers` library has made XLM-R a common baseline and production choice for multilingual NLP.

9.2 Limitations

- **The “curse of multilinguality.”** For a fixed model capacity, adding more languages dilutes the effective per-language capacity, creating a trade-off between the number of supported languages and performance on any individual language.
- **High computational cost.** Pretraining (and even fine-tuning) XLM-R, particularly the Large variant, requires substantial GPU/TPU resources, limiting accessibility for smaller research groups.
- **Large vocabulary and embedding matrix.** The 250K-token vocabulary increases the embedding-layer parameter count substantially, contributing to memory overhead at inference time.
- **Uneven quality across languages.** Despite CC-100’s scale, extremely low-resource or under-digitized languages still receive comparatively little data, and web-crawled text quality varies by language.
- **Encoder-only architecture.** As with BERT/RoBERTa, XLM-R cannot natively perform open-ended text generation; it must be paired with a decoder or adapted (e.g., into an encoder-decoder setup) for generative tasks.
- **Static, context-window-limited input.** Like most Transformer encoders of its generation, XLM-R has a fixed maximum sequence length (512 tokens), which can be limiting for long-document tasks.
- **No explicit cross-lingual alignment signal.** Because XLM-R drops the Translation Language Modeling objective used by XLM, its cross-lingual alignment emerges implicitly from shared subwords and joint training rather than being explicitly supervised, which can be less precise for certain fine-grained cross-lingual tasks.

Chapter 10 Comparative Analysis: BERT, RoBERTa, mBERT, XLM, and XLM-R

10.1 Conceptual Comparison

Table 10.1 summarizes the key distinguishing characteristics of the five closely related models discussed in this report.

Table 10.1: Comparison of BERT, RoBERTa, mBERT, XLM, and XLM-R.

Model	Languages		Pretraining Objective(s)	Training Data	Tokenizer
BERT [3]	English	(monolingual)	MLM + Next Sentence Prediction	BooksCorpus + English Wikipedia (~16 GB)	WordPiece
RoBERTa [4]	English	(monolingual)	MLM only (dynamic masking)	>160 GB English web/book text	Byte-level BPE
mBERT	104 languages		MLM + Next Sentence Prediction	Wikipedia (per-language)	WordPiece (shared, ~110K)
XLM [5]	100 languages		MLM + Translation LM (parallel data)	Wikipedia + parallel corpora	BPE
XLM-R [1]	100 languages		MLM only (monolingual, no parallel data)	CC-100, 2.5 TB filtered Common Crawl	SentencePiece (shared, 250K)

10.2 Discussion

10.2.1 XLM-R vs. BERT and RoBERTa

BERT and RoBERTa are monolingual (English) models. XLM-R adopts RoBERTa’s training recipe (MLM-only, dynamic masking, large batches, long training) but extends it to a multilingual setting. The original paper shows XLM-R Large is competitive with monolingual RoBERTa on the GLUE benchmark, which the authors present as evidence that multilingual training at sufficient scale need not come at the cost of English-language quality.

10.2.2 XLM-R vs. mBERT

mBERT and XLM-R cover a similar number of languages, but differ substantially in training data (small Wikipedia dumps vs. massive filtered Common Crawl), vocabulary size (110K WordPiece vs. 250K SentencePiece), and pretraining objective (mBERT retains Next Sentence Prediction; XLM-R drops it). Empirically, the original paper reports that XLM-R outperforms mBERT by a substantial margin across cross-lingual benchmarks – **+14.6 points average accuracy on XNLI**, **+13 points average F1 on MLQA**, and **+2.4 points F1 on NER**.

10.2.3 XLM-R vs. XLM

The original XLM model relies partly on the Translation Language Modeling (TLM) objective, which requires parallel (translated) sentence pairs – a resource that is expensive and limited for most language pairs. XLM-R shows that dropping TLM entirely, and instead scaling monolingual MLM training data and model capacity, yields better cross-lingual transfer than XLM’s more complex, supervision-dependent recipe.

10.2.4 Summary Table of Reported Gains

Table 10.2 lists the headline improvements XLM-R reports over mBERT, as stated in the original paper’s abstract and results sections.

Table 10.2: Headline performance gains of XLM-R over mBERT, as reported in the original paper.

Benchmark	Improvement over mBERT
XNLI (average accuracy, 15 languages)	+14.6 points
MLQA (average F1)	+13.0 points
CoNLL NER (average F1)	+2.4 points
XNLI, Swahili (vs. earlier XLM)	+15.7 points
XNLI, Urdu (vs. earlier XLM)	+11.4 points

Chapter 11 Applications of XLM-RoBERTa

11.1 General-Purpose Multilingual NLP Tasks

Because it produces high-quality contextual embeddings across 100 languages, XLM-R (typically fine-tuned with a lightweight task-specific head) has been widely applied to:

- **Cross-lingual Natural Language Inference (NLI)** – determining entailment, contradiction, or neutrality between sentence pairs across languages (e.g., the XNLI benchmark).
- **Extractive Question Answering** – locating answer spans within a passage, including cross-lingual QA where the question and passage may be in different languages (e.g., MLQA).
- **Named Entity Recognition (NER)** – identifying and classifying entities (person, organization, location) in text across many languages.
- **Text Classification** – sentiment analysis, topic classification, spam and abuse detection, and misinformation/fake-news detection.
- **Semantic Textual Similarity and Retrieval** – cross-lingual document and sentence retrieval using pooled XLM-R embeddings.
- **Machine Translation Quality Estimation and Data Filtering** – leveraging XLM-R representations to assess translation quality or filter noisy parallel data.
- **Low-resource language processing** – serving as a strong pretrained starting point for languages with very little labeled task-specific data, via transfer learning from high-resource languages.

11.2 Case Study: Bangla Fake News Detection

A particularly illustrative application of XLM-R is **fake news and misinformation detection in Bangla (Bengali)**, a morphologically rich, low-resource-for-NLP language spoken by over 200 million people. Because large labeled Bangla-specific pretrained models were historically scarce, researchers have repeatedly turned to XLM-R as a strong multilingual backbone for this task:

- Studies fine-tuning **XLM-RoBERTa (Base)** with a binary classification head (FAKE / TRUE) on Bangla news datasets report strong classification performance, typically using cross-entropy loss, the AdamW optimizer, and early stopping to prevent overfitting.
- Comparative studies of **BERT, RoBERTa, and XLM-RoBERTa** for Bangla SMS spam detection have found XLM-RoBERTa to be the most effective of the three, outperforming both BERT and RoBERTa on accuracy and F1-score for this Bangla classification task.
- Ensemble approaches that combine XLM-RoBERTa with Bangla-specific models such as **BanglaBERT** and other regional multilingual encoders (e.g., IndicBERT) via soft voting have been proposed to further improve robustness on morphologically rich, code-mixed Bangla text.
- Graph-augmented approaches (e.g., *RoBERTa-GCN*) combine RoBERTa-family contextual embeddings with graph neural networks over news-source relationships to improve fake-news detection specifically for the Bangladeshi news ecosystem.

- Multimodal extensions integrate XLM-R text embeddings with visual features (e.g., from CLIP) to detect misinformation and early warning signs of unrest across combined Bangla and English social-media content.

Why XLM-R fits Bangla fake-news detection: Bangla has comparatively limited labeled NLP resources and relatively few large monolingual pretrained models compared to English. XLM-R’s inclusion of Bangla within its 100-language CC-100 pretraining corpus, combined with its strong zero-/few-shot cross-lingual transfer ability, allows practitioners to fine-tune a single robust multilingual encoder on modest amounts of labeled Bangla news data and still achieve competitive classification accuracy – an efficient alternative to training a Bangla-specific model from scratch.

11.3 Industrial and Production Use

Because of its balance of language coverage and accuracy, XLM-R (particularly the Base variant) is widely used in production multilingual NLP pipelines for content moderation, multilingual search and retrieval, customer-support ticket routing, and multilingual chatbots, where a single deployed model can serve users across many languages without maintaining separate per-language models.

Chapter 12 Performance on Standard Benchmarks

12.1 XNLI: Cross-Lingual Natural Language Inference

XNLI extends the MultiNLI dataset to 15 languages and is the standard benchmark for cross-lingual sentence-pair classification (entailment / contradiction / neutral). Table 12.1 reports representative average cross-lingual zero-shot transfer accuracy figures, as commonly cited from the original paper’s results.

Table 12.1: Approximate XNLI average accuracy (cross-lingual zero-shot transfer, 15 languages), as reported in the original paper.

Model	XNLI Average Accuracy (%)
mBERT	~66.3
XLM (MLM, 100 languages)	~70.2
XLM-R Base	~76.2
XLM-R Large	~80.9
XLM-R Large (translate-train-all)	83.6 (new SOTA at publication)

12.2 MLQA: Cross-Lingual Question Answering

MLQA (Multilingual Question Answering) evaluates extractive QA across seven languages, reporting F1 and Exact Match (EM) scores. The original paper reports that XLM-R achieves an **F1 score of 70.7%** and an **Exact Match score of 52.7%**, exceeding the previous state of the art at the time of publication and improving over mBERT by roughly 13 F1 points on average.

12.3 Named Entity Recognition (NER)

On the CoNLL-2002/2003 NER benchmarks (covering English, Dutch, Spanish, and German), XLM-R improves average F1 score by approximately **+2.4 points** over mBERT, with particularly notable gains on Dutch. Because XLM-R is trained without any task-specific architecture changes, a simple token-classification head fine-tuned on top of the pretrained encoder is sufficient to achieve these results.

12.4 GLUE: Monolingual English Benchmark

Although XLM-R is a multilingual model, the authors also evaluate it on the English-only GLUE benchmark to test whether multilingual training harms English performance. XLM-R Large is shown to be competitive with monolingual RoBERTa on GLUE, supporting the paper’s central claim that multilingual capacity, at sufficient scale, need not come at the expense of per-language quality.

12.5 Summary

Figure 12.1 redraws, in simplified form, the relative ranking of models across the three primary cross-lingual benchmarks discussed above.

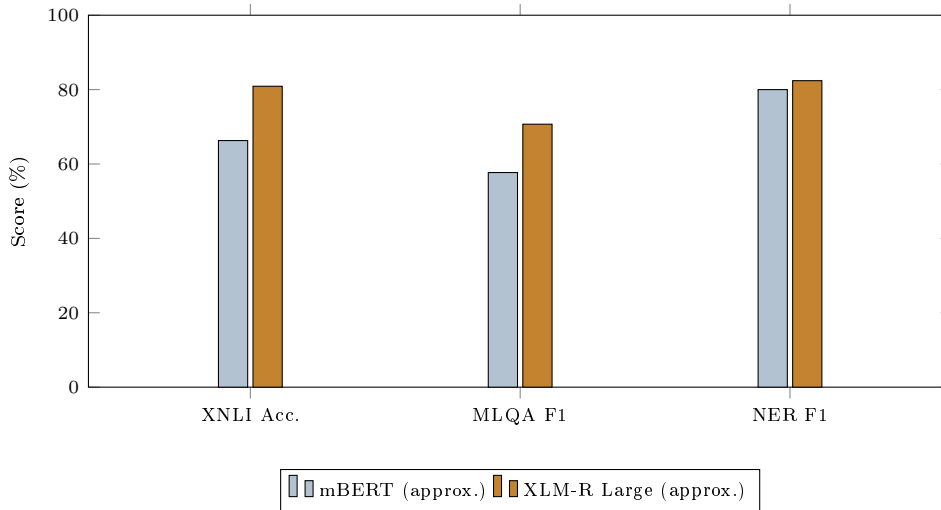


Figure 12.1: Simplified, redrawn comparison of approximate mBERT vs. XLM-R Large scores across three cross-lingual benchmarks. Values are illustrative approximations consistent with the deltas reported in the original paper; consult the original publication for exact per-language figures.

Chapter 13 Summary of the Original Paper

13.1 Citation

Conneau, A., Khandelwal, K., Goyal, N., Chaudhary, V., Wenzek, G., Guzmán, F., Grave, E., Ott, M., Zettlemoyer, L., & Stoyanov, V. (2019/2020). *Unsupervised Cross-lingual Representation Learning at Scale*. arXiv:1911.02116; published at ACL 2020, pp. 8440–8451 [1, 2].

13.2 Motivation

The paper opens from the observation that cross-lingual pretraining had, up to that point, been shown to substantially improve performance on cross-lingual natural language understanding tasks, but prior work (mBERT, XLM) had not carefully studied the effects of scaling such pretraining – in terms of data volume, number of languages, and model capacity. The authors set out to answer: *can massive scale, using only monolingual data and no cross-lingual supervision, produce a single model that is simultaneously strong across many languages, including low-resource ones, without sacrificing performance on high-resource languages?*

13.3 Methodology

- **Model.** A Transformer encoder trained with the Masked Language Modeling objective only (no Next Sentence Prediction, no Translation Language Modeling), following the RoBERTa recipe.
- **Data.** A newly constructed corpus, CC-100, built by filtering Common Crawl snapshots for 100 languages using language identification, deduplication, and LM-based quality filtering, yielding roughly 2.5 TB of text – far larger than the Wikipedia corpora used by mBERT/XLM.
- **Tokenization.** A shared SentencePiece model with a 250K vocabulary trained directly on multilingual raw text, avoiding language-specific preprocessing.
- **Sampling strategy.** An exponential smoothing scheme ($\alpha = 0.3$) balances high- and low-resource language sampling during training.

- **Scaling study.** The authors systematically vary the number of languages, the amount of training data, and model capacity (Base vs. Large) to empirically characterize the trade-off between positive cross-lingual transfer and capacity dilution (the “curse of multilinguality”).

13.4 Experimental Setup

- **Pretraining benchmarks used for evaluation:** XNLI (cross-lingual NLI, 15 languages), MLQA (cross-lingual extractive QA, 7 languages), CoNLL-2002/2003 NER (4 languages), and GLUE (monolingual English understanding).
- **Evaluation protocols:** zero-shot cross-lingual transfer (fine-tune on English only, evaluate on all target languages), translate-train (fine-tune on machine-translated training data per language), and translate-train-all (fine-tune jointly on translated training data from all languages simultaneously).
- **Baselines compared:** mBERT, the original XLM model (in several configurations), and monolingual models such as BERT and RoBERTa (for the English-only GLUE comparison).
- **Model sizes evaluated:** XLM-R Base (270M parameters) and XLM-R Large (550M parameters), each averaged over multiple random seeds for robustness of reported results.

13.5 Results

- XLM-R substantially outperforms mBERT across all cross-lingual benchmarks tested: **+14.6** average accuracy points on XNLI, **+13** average F1 points on MLQA, and **+2.4** F1 points on NER.
- XLM-R shows especially large improvements for **low-resource languages** relative to the earlier XLM model – e.g., **+15.7** points on Swahili and **+11.4** points on Urdu XNLI accuracy – directly attributable to the much larger CC-100 training data for these languages.
- Using the *translate-train-all* protocol, XLM-R Large achieves a new state-of-the-art XNLI average accuracy of **83.6%** at the time of publication.
- On MLQA, XLM-R achieves an F1 of **70.7%** and Exact Match of **52.7%**, exceeding the previous state of the art.
- On the English-only GLUE benchmark, XLM-R Large is shown to be **competitive with monolingual RoBERTa**, demonstrating – for the first time at this scale – that multilingual modeling need not sacrifice per-language performance.
- The scaling study reveals the “**curse of multilinguality**”: for a fixed model capacity, increasing the number of pretraining languages initially improves performance on low-resource languages (positive transfer) but eventually degrades overall performance as capacity is diluted across too many languages – an effect that can be mitigated, but not eliminated, by increasing model size.

13.6 Key Contributions

The paper explicitly claims the following contributions:

1. Demonstrating that multilingual masked language model pretraining can be scaled – in data, languages, and model capacity – to significantly outperform prior cross-lingual models without any cross-lingual supervision (e.g., parallel text).
2. Introducing and releasing the **CC-100** multilingual corpus, filtered from Common Crawl for 100 languages, addressing the data scarcity that limited earlier multilingual models.
3. Providing a systematic empirical analysis of the trade-off between adding more languages and per-language performance (the curse of multilinguality), along with the effect of increasing model capacity to counteract it.
4. Achieving new state-of-the-art results on XNLI, MLQA, and competitive results on NER and GLUE at the time of publication.
5. Showing for the first time that a single multilingual model can be competitive with strong monolingual baselines on English tasks while excelling at cross-lingual transfer.
6. Publicly releasing the CC-100 data pipeline description, pretrained XLM-R models (Base and Large), and code, enabling widespread downstream adoption.

13.7 Key Figures and Tables (Redrawn)

The following redrawn figures summarize, at a conceptual level, the two most-cited empirical results from the paper: the CC-100 vs. Wikipedia data-scale comparison (already presented as Figure 6.1 in Chapter 6) and the XNLI results table (already presented as Table 12.1 in Chapter 12). Figure 13.1 additionally redraws the conceptual shape of the “curse of multilinguality” trade-off described qualitatively in the paper’s scaling analysis.

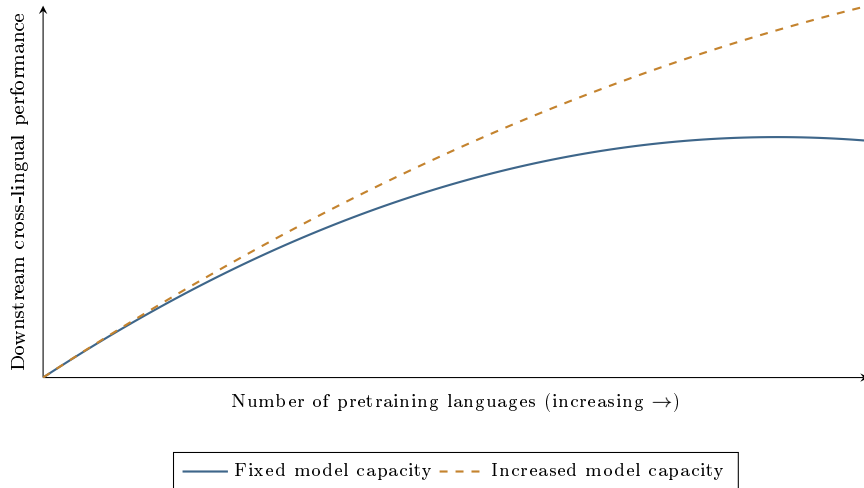


Figure 13.1: Conceptual (redrawn, qualitative) illustration of the “curse of multilinguality”: performance initially rises as more languages are added (positive transfer), then falls as capacity is diluted across too many languages under a fixed model size. Increasing model capacity shifts and flattens this decline. This is a schematic reconstruction of the qualitative trend described in the original paper, not a reproduction of its actual figure or data points.

Table 13.1: Redrawn summary of the original paper’s evaluation protocols and their purpose.

Protocol	Description
Zero-shot cross-lingual transfer	Fine-tune only on English labeled data; evaluate directly on all target languages without further training.
Translate-train	Machine-translate the English training set into each target language; fine-tune a separate model per language on the translated data.
Translate-train-all	Combine machine-translated training data from <i>all</i> target languages into a single joint fine-tuning set, then evaluate on each language.

Chapter 14 Conclusion

XLm-RoBERTa represents a milestone in multilingual Natural Language Processing: it demonstrated that a single Transformer encoder, pretrained with a simple Masked Language Modeling objective over a sufficiently large and clean multilingual corpus (CC-100), can deliver state-of-the-art cross-lingual transfer across 100 languages while remaining competitive with strong monolingual models on English-only benchmarks. By removing the need for parallel/translation data required by earlier cross-lingual models such as XLM, and by scaling training data, vocabulary, and model capacity well beyond mBERT, XLM-R established both a new performance bar on benchmarks such as XNLI, MLQA, and NER, and a widely adopted, freely available foundation model for multilingual downstream applications – from cross-lingual question answering to low-resource text classification tasks such as Bangla fake-news detection. Subsequent work (XLM-R XL/XXL, XLM-V, and various distilled and adapter-based variants) continues to build directly

on the architecture, data pipeline, and empirical findings first established in this paper, underscoring its lasting influence on the field.

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